

The examination is being carried out on the **following application documents**

Description, Pages

1-175 as originally filed

Claims, Numbers

1-5 filed in electronic form on 27-03-2020

Drawings, Sheets

1/8-8/8 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

- D1 WO 2008/039218 A2 (PHARMACYCLICS INC [US]; HONIGBERG LEE [US]; VERNER ERIK [US]; PAN ZHEN) 3 April 2008 (2008-04-03)
- D2 WO 2007/136790 A2 (MANNKIND CORP [US]; FLYNN GARY A [US]; LEE SANDRA AEYOUNG [US]; FARIS) 29 November 2007 (2007-11-29)
- D3 Anna Smith: "Imbruvica fails Phase III pancreatic cancer trial - PharmaTimes",
, 1 January 2019 (2019-01-01), pages 1-2, XP055593391,
Retrieved from the Internet:
URL:http://www.pharmatimes.com/news/imbruvica_fails_phase_iii_pancreatic_cancer_trial_1275758
[retrieved on 2019-06-03]

The following documents are cited by the Examiner. A copy of the documents is annexed to the communication and the numbering will be adhered to in the rest of the procedure.

- D4 Sj Moschos ET AL: "NCI 9922: Phase II study of ibrutinib in treatment-refractory distant metastatic cutaneous melanoma (DMCM)",
Annals of Oncology, Volume 29, Supplement 8, 1 October 2018

(2018-10-01), XP055738183,
DOI: doi:10.1093/annonc/mdy289
[retrieved on 2020-10-08]

D5 W. Gao ET AL: "Selective Antitumor Activity of Ibrutinib in EGFR-Mutant Non-Small Cell Lung Cancer Cells",
JOURNAL OF THE NATIONAL CANCER INSTITUTE,
vol. 106, no. 9, 10 September 2014 (2014-09-10), pages dju204-dju204,
XP055221734,
GB
ISSN: 0027-8874, DOI: 10.1093/jnci/dju204

Art. 83 EPC

The applicant argued with respect to Art. 56 EPC that the cited art failed to provide a motivation to select one of the claimed cancer types, in particular as there was no teaching that BTK inhibition would be successful in treating colon cancer, SCLC, NSCLC or melanoma.

The first examiner is inclined to accept these arguments, i.e. based on the teaching that ibrutinib is an inhibitor of BTK, and to a lesser extent also of SYK and LYN [00412], inhibits growth of DHL-6 cells [00157], inhibits pPLC-gamma and inhibits Ca²⁺ influx in Ramos cells (B-cells), the skilled person might possibly not have deduced that this would translate into an anti-tumor effect in solid tumors as taught in D1 as the **link between these mechanisms of action** and the **claimed tumor types** had not been established in the prior art.

This seems to be further supported by the teaching of D4 (Moschos, et. al., XP55738183) that **ibrutinib failed** to treat melanoma, despite of the fact that patients had been selected that were believed to be particularly susceptible to the drug as they overexpressed ITK. **Thus, even years after the priority date, it is not known, which markers (e.g. overexpression of BTK, ITK, LCK...) were necessary to identify a patient group of cancer patients having a solid cancer, in particular melanoma, that is susceptible to ibrutinib.**

Likewise, D5 (Gao et al. 2014, XP055221734) teaches that NSCLC lung cancer cells tested therein were responsive to ibrutinib, but 36 out of 39 **only at high concentrations, i.e. with IC50 values between 2 and 30 microM**, whereas 3 of them, namely those with mutated EGFR were much more susceptible. Even for the EGFR-

mutated cancer types it was questioned if ibrutinib could be used in clinic. Thus, even years after the priority date, it was still uncertain whether or not ibrutinib would be effective and sufficiently safe to treat EGFR-mutated NSCLC. It is even more doubtful if it would be effective in NSCLC *without* EGFR mutations as covered by the present claims. The application is completely silent as to the specific biomarker to be used in order to identify those NSCLC patients potentially susceptible to ibrutinib in vivo.

The application does not provide evidence that any of the claimed solid cancers are indeed susceptible to ibrutinib. No data have been provided in this respect and no rationale has been provided why the anti-cancer effect as demonstrated for a specific lymphoma type as exemplified in the present application could be extrapolated to other solid tumors.

The ED is aware of the fact that some of ibrutinib's activities have been reported on p. 166/168, tables 1 and 2. However, what has not been disclosed is how these mechanisms are linked with the respective solid tumor. As is apparent from the above cited documents, ibrutinib failed to treat the specific type of melanoma mentioned in Moschos et al., despite of these manifold mechanisms of action and the application does not explain how such failure might be overcome.

Thus, it appears not plausible that the claimed compound treats the claimed solid cancers, at least not over the entire scope of the claims, for the same reasons as why the skilled person would not deduce from D1 that ibrutinib is effective in solid cancers, in particular as the application does not establish which solid cancer is susceptible to which target of ibrutinib and as it requires a research program to identify those solid cancer subtypes susceptible to the compound.

In contrast thereto, the treatment of colon cancer by an EGFR inhibitor was already established and also well-known at the filing date. It appears that the contribution of the present application over the prior art was the finding that ibrutinib was not only effective towards BTK, LYN and Syk, but also towards EGFR. Thus, in as far as the claims are directed to solid cancer types (other than melanoma or NSCLC) that were **commonly known** to be susceptible to EGFR inhibition, such as colon cancer, the subject-matter is considered to be sufficiently disclosed.